

Containerization_and_DevOps

Experiment 11: Orchestration using Docker Compose & Docker Swarm (Continuation of Experiment 6)

PART A – CONCEPT CONTINUATION (Simple Explanation)

From Experiment 6, docker run runs a single container but requires manual handling and has no coordination, while Docker Compose runs multiple containers together but only on a single machine and without auto-healing. The new concept is orchestration, which means automatic management of containers including scaling (increase/decrease containers), self-healing (restart failed containers), load balancing (distribute traffic), and multi-host deployment. The progression path is: docker run → Docker Compose → Docker Swarm → Kubernetes. This experiment focuses on moving from Compose to Swarm.

PART B – PRACTICAL (EXTENSION OF EXPERIMENT 6)

Prerequisites

- Docker installed with Swarm enabled
- docker-compose.yml file (WordPress + MySQL)

```
version: '3.9'
services:
  db:
    image: mysql:5.7
    container_name: wordpress_db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql
  wordpress:
```

```

image: wordpress:latest
container_name: wordpress_app
depends_on:
  - db
ports:
  - "8080:80"
restart: always
environment:
  WORDPRESS_DB_HOST: db:3306
  WORDPRESS_DB_USER: wpuser
  WORDPRESS_DB_PASSWORD: wppass
  WORDPRESS_DB_NAME: wordpress
volumes:
  - wp_data:/var/www/html
volumes:
  db_data:
  wp_data:

```

```

aaroHi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ dir
docker-compose.yml

```

```

GNU nano 7.2 docker-compose.yml
version: '3.9'
services:
  db:
    image: mysql:5.7
    container_name: wordpress_db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    container_name: wordpress_app
    depends_on:
      - db
    ports:
      - "8080:80"
    restart: always
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
    volumes:
      - wp_data:/var/www/html

volumes:
  db_data:
  wp_data:

```

[Read 33 lines]

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo M-6 Copy

Task 1: Check Current State

```

docker compose down -v
docker ps

```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker compose down -v
docker ps
WARN[0000] /mnt/c/Users/ASUS/Experiment11/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
b07bf5fc3d11   sonarqube:lts-community             "/opt/sonarqube/dock...  13 days ago   Up About a minute   0.0.0.0
:9000->9000/tcp, [::]:9000->9000/tcp   sonarqube
eb6cb118b716   postgres:13                         "docker-entrypoint.s..." 13 days ago   Up About a minute   5432/tcp
sonar-db
24f1d004ba3a   containerassgn-mybackend            "uvicorn main:app --..." 4 weeks ago   Up About a minute   0.0.0.0
:8000->8000/tcp, [::]:8000->8000/tcp   containerassgn_mybackend
48bbb6b983c1   containerassgn-mydatabase           "docker-entrypoint.s..." 4 weeks ago   Up About a minute (healthy)
containerassgn_mydatabase
732c618607a6   projectassignment-backend           "docker-entrypoint.s..." 4 weeks ago   Up About a minute   0.0.0.0
:3000->3000/tcp, [::]:3000->3000/tcp   project_node_backend
f98234b377c0   projectassignment-database           "docker-entrypoint.s..." 4 weeks ago   Up About a minute (healthy) 5432/tcp
project_postgres_db
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$
```

Task 2: Initialize Docker Swarm

```
docker swarm init
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker swarm init
Swarm initialized: current node (21qw67z8v81clg7azuasadjwd) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-6bxt2v5od0abgm2atu0mlmnrq4ekzjezacg6tw63qhar9jmaw-0e0jy9i8taew88cmy3hhglrgb 192.168.65.3:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
```

```
docker node ls
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker node ls
ID                HOSTNAME          STATUS    AVAILABILITY    MANAGER STATUS    ENGINE VERSION
21qw67z8v81clg7azuasadjwd *  docker-desktop  Ready     Active           Leader             29.2.0
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$
```

Task 3: Deploy as a Stack

```
docker stack deploy -c docker-compose.yml wpstack
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker stack deploy -c docker-compose.yml wpstack
Ignoring unsupported options: restart

Ignoring deprecated options:

container_name: Setting the container name is not supported.

Since --detach=false was not specified, tasks will be created in the background.
In a future release, --detach=false will become the default.
Creating network wpstack_default
Creating service wpstack_wordpress
Creating service wpstack_db
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$
```

Task 4: Verify Deployment

```
docker service ls
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker service ls
```

ID	NAME	MODE	REPLICAS	IMAGE	PORTS
mqdb8hqimqa6	wpstack_db	replicated	1/1	mysql:5.7	
o97bkdc7ysb	wpstack_wordpress	replicated	1/1	wordpress:latest	*:8080->80/tcp

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$
```

```
docker service ps wpstack_wordpress
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker service ps wpstack_wordpress
```

ID	PORTS	NAME	IMAGE	NODE	DESIRED STATE	CURRENT STATE	ERROR
n9you5nf9g8p		wpstack_wordpress.1	wordpress:latest	docker-desktop	Running	Running 27 seconds ago	

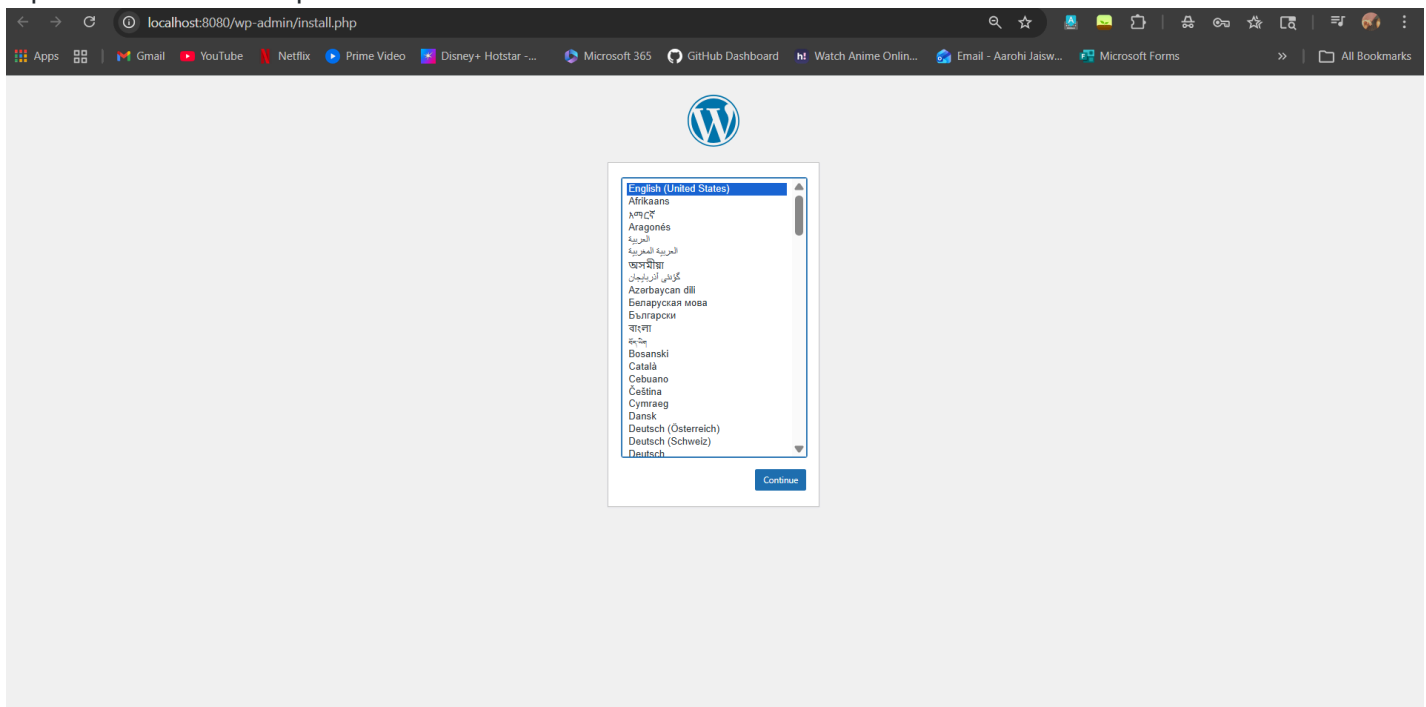
```
docker ps
```

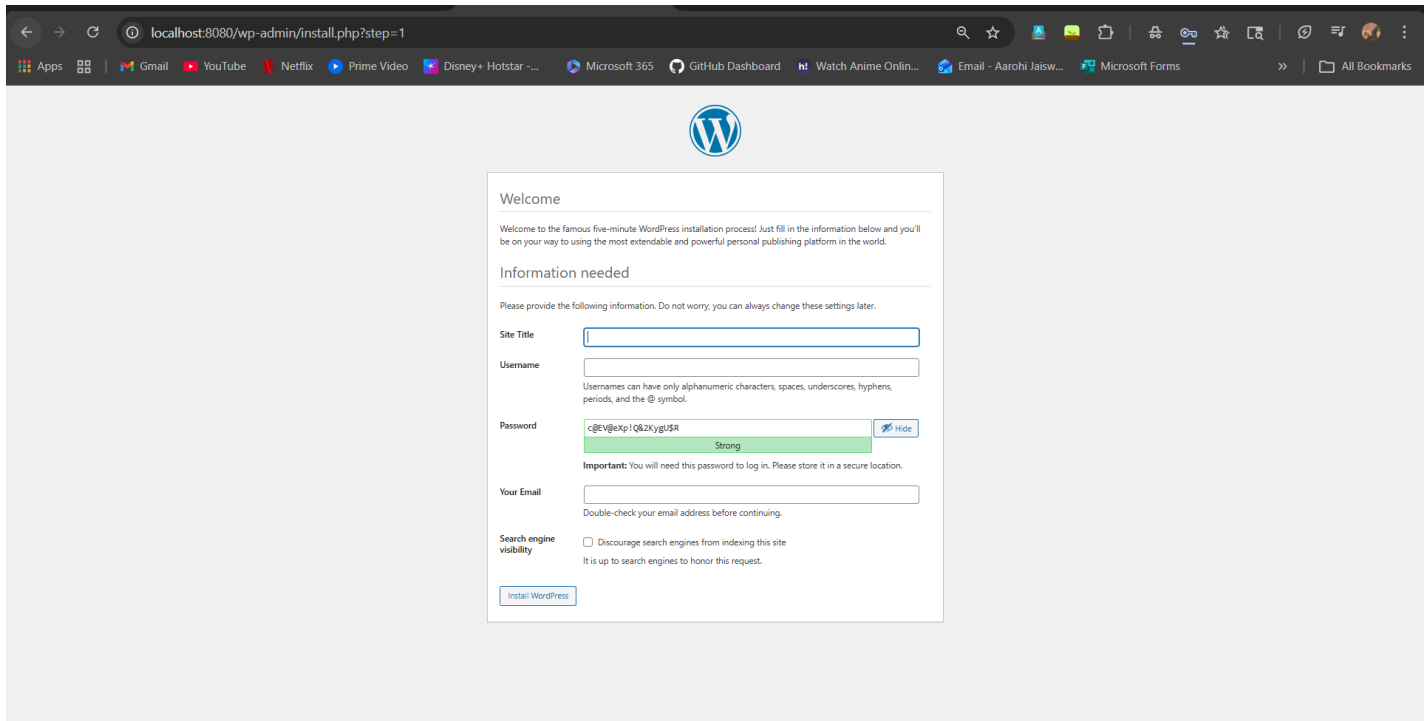
```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker ps
```

CONTAINER ID	IMAGE	NAMES	COMMAND	CREATED	STATUS	PORTS
b8f36f4fa3c6	wordpress:latest		"docker-entrypoint.s..."	2 minutes ago	Up 2 minutes	80/tcp
afb69da49caf	mysql:5.7	wpstack_wordpress.1	"docker-entrypoint.s..."	3 minutes ago	Up 3 minutes	3306/tcp,
33060/tcp		wpstack_db.1	d7hyheo7vl4j1218l6gtulgzl			
b07bf5fc3d11	sonarqube:lts-community		"/opt/sonarqube/dock..."	13 days ago	Up 9 minutes	0.0.0.0:90

Task 5: Access WordPress

Open in browser: <http://localhost:8080>





Task 6: Scale the Application

```
docker service scale wpstack_wordpress=3
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker service scale wpstack_wordpress=3
wpstack_wordpress scaled to 3
overall progress: 3 out of 3 tasks
1/3: running [=====>]
2/3: running [=====>]
3/3: running [=====>]
verify: Service wpstack_wordpress converged
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ |
```

```
docker service ls
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker service ls
ID                NAME                MODE                REPLICAS            IMAGE                PORTS
mqdb8hqimqa6      wpstack_db          replicated          1/1                  mysql:5.7
o97bkdc7ysb       wpstack_wordpress   replicated          3/3                  wordpress:latest     *:8080->80/tcp
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ |
```

```
docker service ps wpstack_wordpress
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker service ps wpstack_wordpress
ID                PORTS                NAME                IMAGE                NODE                DESIRED STATE        CURRENT STATE        ERROR
n9you5nf9g8p      wpstack_wordpress.1  wordpress:latest    docker-desktop       Running              Running 15 minutes ago
ohvplnhag33t      wpstack_wordpress.2  wordpress:latest    docker-desktop       Running              Running 5 minutes ago
epj0t57z58wb      wpstack_wordpress.3  wordpress:latest    docker-desktop       Running              Running 5 minutes ago
```

```
docker ps | grep wordpress
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker ps | grep wordpress
11569571d2c4   wordpress:latest   "docker-entrypoint.s..." 4 minutes ago   Up 4 minutes   80/tcp
6bde4cb3e0c6   wordpress:latest   "docker-entrypoint.s..." 4 minutes ago   Up 4 minutes   80/tcp
b8f36f4fa3c6   wordpress:latest   "docker-entrypoint.s..." 14 minutes ago  Up 14 minutes  80/tcp
```

Task 7: Test Self-Healing

```
docker ps | grep wordpress
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker ps | grep wordpress
11569571d2c4   wordpress:latest   "docker-entrypoint.s..." 4 minutes ago   Up 4 minutes   80/tcp
6bde4cb3e0c6   wordpress:latest   "docker-entrypoint.s..." 4 minutes ago   Up 4 minutes   80/tcp
b8f36f4fa3c6   wordpress:latest   "docker-entrypoint.s..." 14 minutes ago  Up 14 minutes  80/tcp
```

```
docker kill <container-id>
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker kill 11569571d2c4
11569571d2c4
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$
```

```
docker service ps wpstack_wordpress
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker service ps wpstack_wordpress
ID            NAME            IMAGE            NODE            DESIRED STATE   CURRENT STATE           ERROR               PORTS
n9you5nf9g8p  wpstack_wordpress.1  wordpress:latest  docker-desktop  Running         Running 22 minutes ago
ohvplnhag33t  wpstack_wordpress.2  wordpress:latest  docker-desktop  Running         Running 12 minutes ago
y89r0z4mq2jx  wpstack_wordpress.3  wordpress:latest  docker-desktop  Running         Running 17 seconds ago
epj0t57z58wb  wpstack_wordpress.3  wordpress:latest  docker-desktop  Shutdown        Failed 22 seconds ago   "task: non-zero exit (137)"
```

```
docker ps | grep wordpress
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker ps | grep wordpress
638b0c0438f6   wordpress:latest   "docker-entrypoint.s..." About a minute ago   Up About a minute   80/tcp
6bde4cb3e0c6   wordpress:latest   "docker-entrypoint.s..." 13 minutes ago      Up 13 minutes       80/tcp
b8f36f4fa3c6   wordpress:latest   "docker-entrypoint.s..." 23 minutes ago      Up 23 minutes       80/tcp
```

Task 8: Remove the Stack

```
docker stack rm wpstack
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker stack rm wpstack
Removing service wpstack_db
Removing service wpstack_wordpress
Removing network wpstack_default
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ |
```

```
docker service ls
```

```
docker ps
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ docker ps
docker service ls
CONTAINER ID   IMAGE                                COMMAND                                CREATED        STATUS        PORTS                                NAMES
AMES          b07bf5fc3d11   sonarqube:lts-community              "/opt/sonarqube/dock..." 13 days ago   Up 34 minutes   0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp   s
onarqube      eb6cb118b716   postgres:13                          "docker-entrypoint.s..." 13 days ago   Up 34 minutes   5432/tcp                                     s
onar-db       24f1d004ba3a   containerassgn-mybackend             "uvicorn main:app --..." 4 weeks ago   Up 34 minutes   0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp   c
ontainerassgn_mybackend  48bbb6b983c1   containerassgn-mydatabase            "docker-entrypoint.s..." 4 weeks ago   Up 34 minutes (healthy)                                c
ontainerassgn_mydatabase  732c618607a6   projectassignment-backend            "docker-entrypoint.s..." 4 weeks ago   Up 34 minutes   0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp   p
roject_node_backend  f98234b377c0   projectassignment-database           "docker-entrypoint.s..." 4 weeks ago   Up 34 minutes (healthy)   5432/tcp                                     p
project_postgres_db
ID            NAME            MODE            REPLICAS            IMAGE            PORTS
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/Experiment11$ |
```

PART C – ANALYSIS (Compose vs Swarm)

Docker Compose works on a single host with manual recovery and limited scaling. Docker Swarm supports multi-node clusters, automatic scaling, load balancing, and self-healing. Compose is best for development/testing, while Swarm is suitable for small production.

PART D – IMPORTANT OBSERVATIONS

- Same YAML file works for both Compose and Swarm
- docker compose up -d → Compose mode
- docker stack deploy → Swarm mode
- Swarm manages services, not containers
- Built-in load balancer avoids port conflicts

PART E – LEARNING OUTCOME CHECK

- Why is Compose not enough for production?
- Difference between docker compose up and docker stack deploy
- How does Swarm achieve self-healing?
- What happens when a container is killed?
- Can the same Compose file be reused?

SUMMARY

You progressed from single containers to multi-container applications and now to orchestration using Docker Swarm. You learned scaling, self-healing, and load balancing. Compose defines the application, Swarm runs it reliably.

QUICK REFERENCE COMMANDS

```
docker swarm init
docker stack deploy -c docker-compose.yml <stack-name>
docker service ls
docker service scale <stack-name_service-name>=<replicas>
docker service ps <service-name>
docker stack rm <stack-name>
docker swarm leave --force
```